

Eklavya Sharma

Curriculum Vitae

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in [sharmaeklavya2](#)  [sharmaeklavya2](#) 

Research Interests

Economics and computation, Social choice theory, Game theory, Markets, Approximation algorithms

Education

- Aug 2021 – **PhD**, *Department of Industrial & Enterprise Systems Engineering (ISE)*,
present *University of Illinois at Urbana-Champaign (UIUC)*, IL, USA
Research on data markets and fair allocation. Advised by Prof. Jugal Garg .
- July 2019 – **M.Tech. (Research)**, *Computer Science and Automation (CSA)*, Indian
July 2021 *Institute of Science (IISc)*, Bangalore, GPA: 9.7 / 10.0
Research on approximation algorithms for variants of bin packing and knapsack.
Advised by Prof. Arindam Khan .
- Aug 2014 – **B.E. (Hons) Computer Science**, *Birla Institute of Technology and Science*
June 2018 *(BITS)*, Pilani, India, GPA: 9.14 / 10.00

Publications

- Submitted to Equilibrium pricing for oligopolistic data markets
ICML'26 with Bhaskar Ray Chaudhury, Jugal Garg, and Jiaxin Song
- Submitted to Revenue-optimal pricing for budget-constrained buyers in data markets
STOC'26 with Bhaskar Ray Chaudhury, Jugal Garg, and Jiaxin Song
- AAMAS'26 Exploring relations among fairness notions in discrete fair division 
with Jugal Garg
- AAMAS'26 Proportional and Pareto-optimal allocation of chores with subsidy 
with Jugal Garg and Xiaowei Wu
- EC'24 Improving approximation guarantees for maximin share  
with Hannaneh Akrami, Jugal Garg, and Setareh Taki
- IJCAI'23 New fairness concepts for allocating indivisible items  
with Ioannis Caragiannis, Jugal Garg, Nidhi Rathi, and Giovanna Varricchio
- IJCAI'23 Simplification and improvement of MMS approximation  
with Hannaneh Akrami, Jugal Garg, and Setareh Taki

- FSTTCS'23 Two-player matrix games repeated until collision [↗](#)
with Aniket Murhekar
- ArXiv Automating the search for small hard examples to approximation algorithms [↗](#)
- FSTTCS'22 Approximation algorithms for multidimensional packing [↗](#) [↗](#)
with Arindam Khan and K.V.N. Sreenivas
- Algorithmica, APPROX'21 Tight approximation algorithms for geometric bin packing with skewed items [↗](#) [↗](#) [↗](#), with Arindam Khan
- FSTTCS'21 Harmonic algorithms for packing d -dimensional cuboids into bins [↗](#) [↗](#)
- ICACCI'18 Mitigating DNS amplification attacks using a set of geographically distributed SDN routers [↗](#), with Vishal Gupta

Fellowships and Achievements

- Aug 2024 **Mavis Future Faculty Fellowship, UIUC** [↗](#)
A program that trains fellows on various aspects of an academic career through workshops, seminars, and activities.
- April 2024 **William A. Chittenden II Award, UIUC** [↗](#)
Given to an outstanding graduate student at ISE.
- July 2023 **Dr. MNS Swamy Medal for Best MTech (Research) Thesis, IISc** [↗](#)
- April 2023 **Sharp Outstanding Graduate Student Award, UIUC** [↗](#)
- Aug 2021 – July 2022 **Samuel Brainin Engineering Fellowship, UIUC**
- March 2018 **Graduate Aptitude Test in Engineering (GATE), India**
Rank 86 (out of $\sim 100,000$ candidates) in the 'Computer Science and IT' test.
- Aug 2014 – Dec 2015 **BITS Pilani Merit Scholarship**
Scored GPA among top 2% of students in three semesters.
- ACM-ICPC** [↗](#)
ACM-ICPC is an international annual multi-tiered programming contest for college students. Around 3000 teams (of 3 students each) participate in the Indian online qualifying round each year. Top few teams qualify for on-site regional contests in India.
- Dec 2017 Ranked 29 out of 250 teams in the Amritapuri regional contest.
- Dec 2016 Ranked 66 out of 450 teams in the Amritapuri regional contest.
- Dec 2016 Ranked 30 out of 70 teams in the Kharagpur regional contest.
- Dec 2015 Ranked 88 out of 250 teams in the Amritapuri regional contest.

Invited Talks

- 27 Oct 2023 **Fair allocation of indivisible items**
Capital Area Theory Seminar, University of Maryland, College Park

22 Dec 2022 **Existence and computation of epistemic EFX allocations** 
Indian Institute of Science, Bangalore

Professional Service

Program committee member for AAMAS'26, AAI'26, EC'25, AAMAS'25, AAMAS'24.

Conference subreviewer for EC'24, ICALP'24, EC'23, ESA'23, IJCAI'23, FCT'23, STOC'22, SAGT'22, MFCS'21.

Journal reviewer for Optimization Letters (2024).

Implementation Projects

April 2024 – **Exploring relations among fairness notions in discrete fair division**
Dec 2024 *Published in AAMAS'26, code at github.com/sharmaeklavya2/cpigjs*

Wrote a program in JavaScript that helped prove many new results for the problem of fairly allocating indivisible items among multiple agents. These results help arrange fairness notions in a hierarchy.

Sept 2023 – **Automating the search for hard examples to approximation algorithms.** *code: github.com/sharmaeklavya2/code2dtree, paper in IJCAI'23.*
March 2024

Wrote a python library that converts any function to a decision tree (for a fixed input size). Then wrote a program to find tight hard examples for approximation algorithms by running a linear program for each leaf of the algorithm's decision tree. Used this to find tight hard examples for the then-best-known approximate-MMS algorithm for fairly allocating goods.

May 2020 **Interactive app for rectangular bin packing**
code: github.com/sharmaeklavya2/packing-game

Work Experience

Teaching Assistant, IE 300: Analysis of Data (Fall'22, Spring'24, Spring'25), UIUC

Spring 2023 **Teaching Assistant, IE 310: Deterministic models in optimization**, UIUC

Fall 2020 **Teaching Assistant, Design and Analysis of Algorithms**, IISc Bangalore

Aug 2018 – **Software Engineer, media.net**, Bangalore, India

July 2019 *Topics: machine learning, large-scale systems.*

media.net is an advertisement-technology company. I worked on improving their machine-learning-based algorithm for bidding in real-time ad auctions.

Jan 2018 – **Intern** , American Express, Gurgaon, India

June 2018 *Topics: neural networks, machine learning, big data.*

Trained a neural network to predict credit-card defaulting. The input format was unconventional, so a custom architecture was devised. Its performance was at par with the production model, which was tuned over many years.

May 2017 – **Intern**, *Directi*, Mumbai, India

July 2017 *Topics*: machine learning.

Made Directi's news article classification algorithm recognize more categories.

May 2016 – **Google Summer of Code (GSoC) Student** ↗, *Zulip*

Aug 2016 *Topics*: software development.

Zulip is an open-source group chat application. 3 students were selected from over 100 applicants to work on Zulip as part of the GSoC program.

- Annotated python code ($\sim 50,000$ lines) for use with a static type-checker.
- Migrated code to Python 3 by switching to newer dependencies, using automated code conversion, standardizing string types, and fixing bugs.

Selected Coursework

UIUC:

- (CS 580) Algorithmic Game Theory: grade A
- (CS 598 TH1) Recent Advances in Theoretical CS: grade A+
- (CS 473) Algorithms: grade A+
- (IE 511) Integer Programming: grade A
- (IE 519) Combinatorial Optimization: grade A
- (IE 410) Advanced Stochastic Processes and Applications: grade A+
- (IE 411) Optimization of Large Systems: grade A+

IISc Bangalore:

- Approximation Algorithms: grade A+, rank 1
- Design and Analysis of Algorithms: grade A+, rank 1
- Computational Methods of Optimization: grade A+, rank 1
- Cryptography: grade A

Computer Skills

L^AT_EX, Python, HTML, CSS, JavaScript, C/C++, Java, SQL.